

Unit #7

Lines and Planes in 3-Space

EQUATIONS OF LINES IN TWO SPACE:

In order to determine a straight line it is enough to specify either:

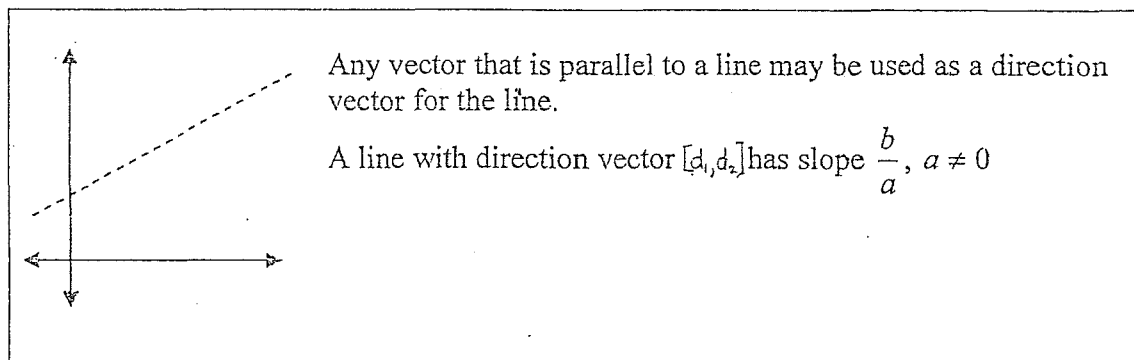
1.

OR

2.

For a line, l , a fixed vector, $\vec{d}$, is called a **direction vector** for the line if it is parallel to l .

Note: every line has an infinite number of direction vectors that can be represented as $t\vec{d}$ where $\vec{d}$ is one direction vector for the line and t is a non-zero real number.



Example: Find a direction vector for each line:

a) The line l_1 through points $A(4, -5)$ and $B(3, -7)$.

b) The line l_2 with slope $\frac{4}{5}$.

The **Vector Equation of the Line** through the point of the line with position vector, $\vec{r}_1$, in the direction of $\vec{d}$ is of the form:

Example: Find vector equations of the line through points $A(1,7)$ and $B(4,0)$.

The real number, t , is called a **parameter**, each number t specifies one point on the line and to each point in the line there corresponds one value of t .

Example: Find three points on the line with vector equation: $\vec{p} = [0,4] + t[5,-2]$

The **Parametric Equations of the Line** passing through (x_1, y_1) in direction $[d_1, d_2]$ are of the form:

Example: Find parametric equations of the straight line through the point $P_1(1,2)$ and having a direction vector $\vec{d} = [3,1]$.

A Symmetric Equation of the Line through (x_1, y_1) with direction vector (a, b) is

Example: Find a symmetric equation of the line through points $A(2, -3)$ and $B(1, 0)$.

Example: Are the lines by the following vector equations coincident? That is, do these equations represent the same straight line?

$$\vec{r} = (3, 4) + s(2, -1) \quad \text{and} \quad \vec{r} = (-9, 10) + t(-6, 3)$$

EQUATIONS OF LINES IN THREE SPACE:

Vector Equations of the Line through the point with position vector $\vec{r}_1$, in direction $\vec{d}$ is of the form:

Parametric Equations of the Line through the point (x_1, y_1, z_1) in the direction (a, b, c) is of the form:

Symmetric Equations of the Line through (x_1, y_1, z_1) with direction (a, b, c) is of the form:

Example: Find the vector, parametric and symmetric equations of the line through the points $P_1(3, 2, 4)$ and $P_2(-1, 7, 2)$. Does the $Q(-3, 5, 2)$ lie on this line?

Intersection of 2 Lines

1. Find the vector equation of a line passing through $(3,0,2)$ and:

a) perpendicular to both $l_1 : (x, y, z) = (1, 2, 3) + k(4, -1, 2)$

$$l_2 : (x, y, z) = (2, -1, 4) + m(1, 0, -3)$$

b) parallel to the line $(x, y, z) = (-2, 6, 7) + t(-2, 1, 3)$

2. Find three points on the line $(x, y, z) = (-5, 3, 7) + s(2, -1, 4)$

3. Are $\frac{x-4}{2} = y-6 = z-5$ and $(x, y, z) = (6, 8, 7) + k(-4, -2, -2)$ the same line?

Example 1: Show that the planes are parallel and distinct.

$$\pi_1 : x + 2y + 3z - 6 = 0$$

$$\pi_2 : 4x + 8y + 12z - 25 = 0$$

Example 2: Show that the planes are parallel and identical.

$$\pi_1 : 3x + 2y + 5z = 4$$

$$\pi_2 : 9x + 6y + 15z - 12 = 0$$

Intersection of three planes

Possibilities:

- Planes are parallel and distinct – no solution (normals are collinear)
- If only one pair of normals is collinear, then there is a pair of parallel planes
- If the normals are coplanar, but no pair is collinear, then either the planes have no point in common or they have a line in common
- If the normals are non-coplanar, then the planes intersect in a single point

Determine the intersection, if any, of each of the following sets of planes. In each case, give a geometrical interpretation of the system of equations and the solution. Also, state whether the system has no solution, a unique solution or an infinite number of solutions.

$$\pi_1 : 3x - 2y + 5z = 8$$

$$1) \pi_2 : 6x - 4y + 10z = -7$$

$$\pi_3 : 24x - 16y + 40z = -1$$

$$\pi_1 : 3x + 5y - 2z - 4 = 0$$

$$2) \pi_2 : 4x - 7y + 6z + 11 = 0$$

$$\pi_3 : 6x + 10y - 4z + 1 = 0$$

$$\pi_1 : x + 2y + 3z - 4 = 0$$

$$3) \pi_2 : 2x - y + 4z + 7 = 0$$

$$\pi_3 : 3x - 14y + z - 31 = 0$$

$$\pi_1 : x + y - 3 = 0$$

$$4) \pi_2 : y + z + 5 = 0$$

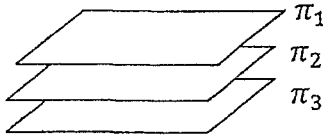
$$\pi_3 : x + z + 2 = 0$$

Intersection Of Three Planes

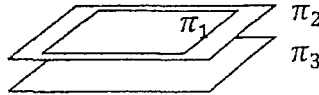
*****Major Step:** Determine how many normal vectors are parallel to each other***

Case 1: Three normals are parallel to each other. i.e. $\vec{n}_1 = k\vec{n}_2 = s\vec{n}_3$, $k, s \in \mathbb{R}$

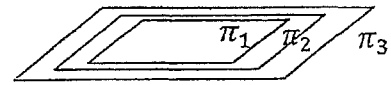
- Therefore, we have 3 parallel planes.



No pair of D values
are scalar multiples



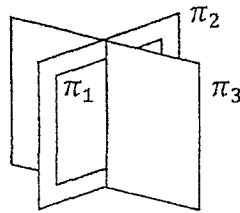
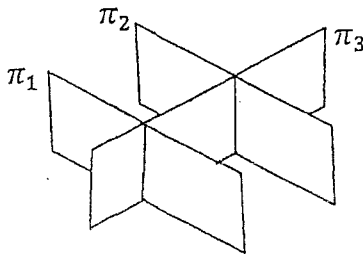
One pair of D values
are scalar multiples
 $D_1 = kD_2$



Two pairs of D values
are scalar multiples
 $D_1 = kD_2 = sD_3$

Case 2: Two normals are parallel to each other.

- Therefore, we have 2 parallel planes



Check to see if two planes are coincident.

- If not, you have the first picture
 - No Intersection
- If so, you have the second picture
 - Planes intersect in a line

Note: For the second picture:

$$\vec{n}_1 = k\vec{n}_2 \text{ and } D_1 = kD_2$$

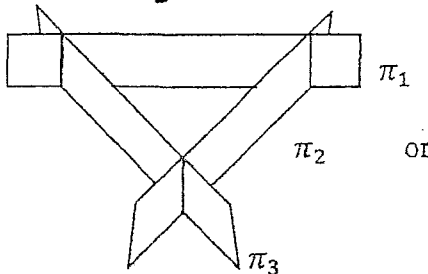
Case 3: No normals are parallel to each other

- Therefore, we have 0 parallel planes

→ Check if normals are coplanar i.e. Perform $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) = 0$??? or **Directly Solve**

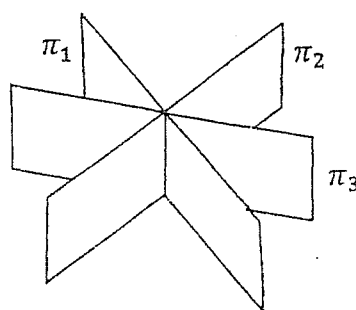
If normals are coplanar (i.e. $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) = 0$) you have:

Triangular Prism



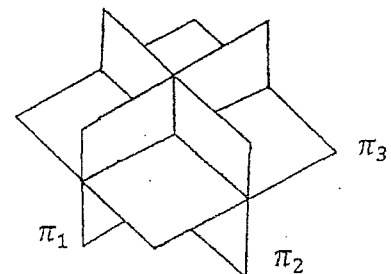
No intersection

Pinwheel



Line of intersection

If normals are not coplanar (i.e. $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) \neq 0$) you have:



One point of intersection

→ You must solve the system to determine which case.

Strategy: Use ① and ② to get ④, then ① and ③ to get ⑤. Use ④ and ⑤ to solve for variables.

Find: 1) Contradiction ($0=3$) 2) True Statement ($0=0$) or 3) Unique values for variables

INTERSECTION OF THREE PLANES

CASE 1: THREE PARALLEL PLANES

- 1) The three planes are parallel and distinct. They do not intersect.
- 2) There are three parallel planes. Two are parallel coincident and one is distinct. The planes do not intersect
- 3) The three planes are parallel and coincident. They intersect in the plane.

CASE 2: TWO PARALLEL PLANES AND ONE NON-PARALLEL

- 1) There are two parallel distinct planes and one non-parallel plane. The three planes do not intersect.
- 2) There are two parallel coincident planes and one non-parallel plane. The three planes intersect in a line.

CASE 3: THREE NON-PARALLEL PLANES

A: THE NORMALS ARE COPLANAR

- 1) The three planes do not intersect. They form a triangular prism.
- 2) The three planes intersect in a line.

B: THE NORMALS ARE NOT COPLANAR

The three planes intersect in a point.

SUMMARY OF LINES AND PLANES

- 1) In three-space, there are vector equations of a line. For each vector equation, there is a set of parametric equations and symmetric equations. A line has many vector equations.
- 2) Two lines can be parallel distinct, parallel coincident or non-parallel. Non-parallel lines intersect in a point or are skew lines and do not intersect.
- 3) The equation of a plane can be expressed in vector form, parametric form and scalar form. Only the scalar form of the equation is unique.
- 4) A line and a plane can intersect in the line or a point; otherwise, the line and the plane are parallel distinct and do not intersect.
- 5) Two non-parallel planes intersect in a line. Two parallel distinct planes do not intersect. Two parallel coincident planes intersect in the plane.
- 6) Three parallel coincident planes intersect in the plane. If not all three are coincident, then the three parallel planes do not intersect.
- 7) Two parallel coincident planes and one non-parallel plane intersect in a line.
- 8) Two parallel distinct planes and one non-parallel plane do not intersect.
- 9) If the normals are coplanar for three non-parallel planes, then the three planes intersect in a line or form a triangular prism and do not intersect.
- 10) If the normals are not coplanar for three non-parallel planes, then the three planes intersect in a point.

- 1) Find vector and parametric equations of the line that is parallel to $\frac{x-5}{4} = \frac{y+2}{-2} = \frac{z+6}{5}$ and passes through the point $(0, 6, 0)$
- 2) Given the line $\vec{r} = [12, -8, -4] + t[-3, 4, 2]$, find the intersection(s) with the xy plane
- 3) Find the shortest distance between the point $(1, 3, 2)$ and the line $\frac{x-1}{-1} = \frac{y-3}{1} = \frac{z-7}{2}$
- 4) Find vector and parametric equations of the plane that contains the two intersecting lines $\vec{r}_1 = [3, -1, 2] + s[4, 0, 1]$ and $\vec{r}_2 = [3, -1, 2] + t[4, 0, 2]$
- 5) Find a scalar equation of the plane that contains the origin and the point $(2, -3, 2)$ and is perpendicular to the plane $x + 2y - z + 3 = 0$.
- 6) For what values of k will the planes $2x - 6y + 4z + 3 = 0$ and $3x - 9y + 6z + k = 0$ intersect?
- 7) Find the equation of the plane that passes through the point $(3, 0, -4)$ and is perpendicular to the line of intersection of the planes $x + 2y - 7z - 3 = 0$ and $x - 5y + 4z - 1 = 0$
- 8) Solve the following system of equations. Give a geometrical interpretation of the system and its solution.
$$\begin{aligned}x + 3y + 3z &= 8 \\x - y + 3z &= 4 \\2x + 6y + 6z &= 16\end{aligned}$$
- 9) Are these two planes parallel, coincident, or neither?
$$\begin{aligned}\vec{r} &= [4, 0, 3] + s[-8, 1, -9] + t[-1, 5, 7] \\ \vec{p} &= [-14, 12, -1] + u[1, 1, 3] + v[-2, 1, -1]\end{aligned}$$

Unit 7: 3D lines and Planes Review Questions

- Is the following statement true or false? Explain. "In order to find the equation of a plane, all you need are 3 points that lie on that plane" [sometimes true]
- Find the scalar equation of the plane that goes through the points $A(1, -1, 3)$, $B(0, 1, 7)$, and $C(4, 0, -1)$.
[π : $-12x + 8y - 7z + 41 = 0$]
- Write the scalar equation of a plane that passes through $A(-1, 1, 3)$ and also contains the line $[x, y, z] = (4, 1, 2) + t[-2, 1, 3]$.
[π : $x - 13y + 5z - 1 = 0$]
- Find the equation of the plane that contains $l_1: [x, y, z] = (1, -2, 1) + t[3, 1, -1]$ and is perpendicular to $l_2: \frac{x-2}{2} = \frac{y+7}{-3} = \frac{z}{3}$.
[π : $2x - 3y + 3z - 11 = 0$]
- Find the vector and scalar equation of the plane that contains the lines $l_1: [x, y, z] = (2, 3, -1) + t[10, 12, 16]$ and $l_2: [x, y, z] = (-1, -2, 1) + k[5, 6, 8]$.
[π : $[x, y, z] = (2, 3, -1) + t[5, 6, 8] + k[-3, -5, 2]$ [π : $52x - 34y - 7z - 9 = 0$]
- Find the intersection of the following lines $l_1: \frac{x-3}{1} = \frac{y+7}{-2} = \frac{5-z}{-4}$ and $l_2: \frac{x+7}{3} = \frac{y+8}{1} = \frac{-z+4}{1}$ [$2, -5, 1$]
- Find the intersection of $l_1: [x, y, z] = (0, 1, -4) + t[2, -1, 1]$ and $\pi_1: x + 4y + 2z - 4 = 0$ if it exists.
[DNE]
- Find the distance from point $A(1, 2, 3)$ to the plane $\pi_1: 2x - 3y + 6z = 5$ [$\frac{9}{7}$]
- Determine the distance between the two given skew lines $l_1: [x, y, z] = (3, -7, 5) + k[1, -2, 4]$ and $l_2: [x, y, z] = (-7, -8, 0) + m[3, 1, -1]$ [$\frac{28}{\sqrt{222}}$]
- Use normals to analyze and describe the geometric relationship of the given set of planes.
Give reasons for your description. Do not solve. $\begin{cases} \pi_1: x + y - 6z = -1 \\ \pi_2: 4x - 3y + 2z = 9 \\ \pi_3: 2x + 9y = 4 \end{cases}$ [pt]
- Solve $\begin{cases} \pi_1: 6x - 3y + 9z - 6 = 0 \\ \pi_2: 4x - 2y + 6z - 3 = 0 \\ \pi_3: x - 3y + 2z + 10 = 0 \end{cases}$ [2 parallel distinct planes, one non parallel $\therefore$ no intersection]
- Solve $\begin{cases} \pi_1: 3x - 3y - 2z - 14 = 0 \\ \pi_2: 5x + y - 6z - 10 = 0 \\ \pi_3: x - 2y + 4z - 9 = 0 \end{cases}$ [Pol: $(3, -2, \frac{1}{2})$]

Review Questions

12. Find a vector equation, parametric equations, and symmetric equations for each of the following lines.

- through the point $A(-2, 3, 0)$ with direction vector $\langle 4, 2, -2 \rangle$
- through the points $A(-2, 3, 5)$ and $B(3, -1, 6)$
- through the point $A(1, 0, 10)$ with direction numbers 1, 2, and -4
- through the point $B(5, 6, 10)$, parallel to the line $\vec{r} = \langle 6, 1, 1 \rangle + k\langle -2, 1, 3 \rangle$
- through the point $C(-4, 3, 6)$ and parallel to the x-axis
- through the point $C(-4, 3, 6)$ and parallel to the z-axis

13. In each of the following, determine whether or not the given point lies on the line with the given equation.

- | point | line equation |
|-------------------|---|
| a) $(-1, 4, 9)$ | $\vec{r} = \langle 1, 3, 4 \rangle + k\langle -2, 1, 5 \rangle$ |
| b) $(-10, -8, 4)$ | $\vec{r} = \langle 0, 1, 0 \rangle + k\langle 5, 4, -1 \rangle$ |
| c) $(11, 8, 0)$ | $\vec{r} = \langle 5 + 2k, -4 + 4k, 3 - k \rangle$ |

14. a) Find a vector equation of the line that passes through the points $A(3, -2, 1)$ and $B(4, 0, 7)$.

b) Show that the line with equation $\vec{r} = \langle 5, 2, 13 \rangle + t\langle 1, 2, 6 \rangle$ is the same as the line in part a).

15. Find a vector equation of the line that passes through the point $A(-1, -2, 4)$ and is parallel to the line with equation

$$\vec{r} = \langle 0, 4, 9 \rangle + k\langle -3, 1, 4 \rangle.$$

16. Find a vector equation of the line passing through the point $A(4, 1, 5)$ that is perpendicular to vector $\vec{u} = \langle 0, -1, 7 \rangle$ and also perpendicular to vector $\vec{v} = \langle 2, 4, -3 \rangle$.

17. Find a vector equation of the line with symmetric equations

$$\frac{x-3}{4} = \frac{y+1}{5} = \frac{z+3}{-1}$$

18. Find the value of t so that the two lines

$$\vec{r} = \langle 1, 7, 3 \rangle + k\langle -2, 3, 5t \rangle \text{ and}$$

$$\vec{r} = \langle 4, 1, 2 \rangle + s\langle t, 1, -3 \rangle \text{ will be perpendicular.}$$

19. Find the value of the variable k so that the line with symmetric equations

$$\frac{x-3}{2} = \frac{y+1}{4} = \frac{z+3}{k} \text{ is perpendicular to the}$$

line with vector equation

$$\vec{r} = \langle 5 + 2ks, 3 - 4s, -8 - 12s \rangle$$

20. Find the value of the variable t so that the following lines are perpendicular.

$$\frac{x-3}{3t+1} = \frac{y+6}{2} = \frac{z+3}{2t}, \quad \frac{x+7}{3} = \frac{y+8}{-2t} = \frac{z+9}{-3}$$

34. For each of the following pairs of lines in 3-space determine whether or not the lines are parallel. If the lines are not parallel, determine whether the lines intersect or are skew. If the lines intersect, find the point of intersection.

a) $\vec{r} = \langle 3 + k, 7 - 2k, 4 + 3k \rangle$

$$\vec{r} = \langle -2 + 2t, 1 + 4t, 11 - 5t \rangle$$

b) $\vec{r} = \langle 2 + 3k, 1 + 4k, 4 + k \rangle$

$$\vec{r} = \langle 3 + b, 1 + 2b, -2 + b \rangle$$

c) $\vec{r} = \langle -1 + a, 4 - 3a, -6 + a \rangle$

$$\vec{r} = \langle 2 - 3t, -5 + 9t, -3 - 3t \rangle$$

d) $\vec{r} = \langle 6 - 2w, -9 + 4w, 7 + w \rangle$

$$\vec{r} = \langle 7 - 5k, -3 + 2k, -3 - 4k \rangle$$

36. Find the point of intersection of the line

$$\vec{r} = \langle 0, -8, 4 \rangle + k\langle 2, 7, -1 \rangle \text{ and the line}$$

through the points $A(8, 4, -1)$ and $B(0, 8, 5)$.

37. Find the point of intersection of the lines

$$\frac{x+1}{4} = \frac{y+2}{-2} = \frac{z}{3}, \quad \frac{x+7}{5} = \frac{y+10}{3} = \frac{z-5}{-1}$$

38. a) Prove that the lines

$$\vec{r} = \langle 1 + k, 4 + 3k, 3 - 2k \rangle \text{ and}$$

$$\vec{r} = \langle -3 + 3t, 6 + 2t, -3t \rangle \text{ are skew lines.}$$

b) Find the distance between the given skew lines.

39. Given the lines $\vec{r} = \langle -1 + k, 3 + 2k, 1 - 3k \rangle$

$$\text{and } \vec{r} = \langle 3 + 2w, 10 + 3w, -6 - w \rangle.$$

a) Prove that the lines intersect.

b) Find the angle between the lines.

(ANSWERS)

12. a) $r = (2, 5, 0) + k(4, 2, -2)$

$$x = 2 + 4k$$

$$y = 5 + 2k$$

$$z = -2k$$

$$\frac{x-2}{4} = \frac{y-5}{2} = \frac{z}{-2}$$

b) $r = (-2, 3, 5) + t(5, -4, 1)$

$$x = -2 + 5t$$

$$y = 3 - 4t$$

$$z = 5 + t$$

$$\frac{x+2}{5} = \frac{y-3}{-4} = \frac{z-5}{1}$$

c) $r = (1, 0, 10) + s(1, 2, -4)$

$$x = 1 + s$$

$$y = 2s$$

$$z = 10 - 4s$$

$$\frac{x-1}{1} = \frac{y}{2} = \frac{z-10}{-4}$$

d) $r = (5, 6, 10) + m(-2, 1, 3)$

$$x = 5 - 2m$$

$$y = 6 + m$$

$$z = 10 + 3m$$

$$\frac{x-5}{-2} = \frac{y-6}{1} = \frac{z-10}{3}$$

e) $r = (-4, 3, 6) + k(1, 0, 0)$

$$x = -4 + k$$

$$y = 3$$

$$z = 6$$

none exists

f) $r = (-4, 3, 6) + t(0, 0, 1)$

$$k = -4$$

$$y = 3$$

$$z = 6 + t$$

none exists

3. a) yes

b) no

c) yes

4. a) $r = (3, -2, 1) + k(1, 2, 6)$

5. $r = (-1, -2, 4) + s(-3, 1, 4)$

6. $r = (4, 1, 5) + k(-25, 14, 2)$

17. $r = (3, -1, 3) + k(4, 5, -1)$

18. $\frac{3}{17}$

19. -2

20. 3

34. a) intersect at (2, 9, 1)

b) skew

c) parallel and distinct

d) skew

35. a) consistent and independent

b) inconsistent

c) inconsistent

d) inconsistent

36. (4, 6, 2)

37. (3, -4, 3)

38. b) $\frac{35}{\sqrt{83}}$

39. a) (1, 7, -5)

b) 38°